Python 3

BEGINNER

COURSE SLIDES · NON-WSQ COURSE

Python Fundamentals Masterclass

Course Code: C179

Duration: 2 days · 15 instructional hours (9:30am – 5:30pm)
Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome & Housekeeping

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About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

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YOUR TRAINER

About the Trainer



Dr Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte Ltd

ROLE
Principal Trainer and Founder, Tertiary Infotech Academy Pte Ltd

QUALIFICATIONS
PhD · MSc · ACTA/DACE-certified adult educator

EXPERTISE
Python, data analytics, machine learning and AI applications

DELIVERS
Courses on Python programming, data science and machine learning

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ICE-BREAKER

Let's Know Each Other

- Your name and organisation / role.
- Your experience with programming or data analysis (if any).
- What you want to automate or analyse with Python after this course.

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HOUSEKEEPING

Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively — no question is stupid.
3. Mutual respect: agree to disagree.
4. One conversation at one time.
5. Be punctual; return from breaks on time.
6. Full attendance — attend both days to get the most from the course.

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LMS / TMS · lms-tms.tertiaryinfotech.com

Download Your Course Material

1. Open lms-tms.tertiaryinfotech.com in your browser
2. Log in with your registered email
3. Open your course: Python Fundamentals Masterclass
4. Open the Course Materials section
5. Download the Slides and Learner Guide (PDF)

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SCHEDULE

Lesson Plan — 2 Days, 7.5 hours/day

Day 1

- Day 1 — Python Basics — Data Types, Operators, Control Structures
  - Welcome & introductions
  - Topic 1: Get Started on Python
  - Topic 2: Data Types
  - Lunch (30 min)
  - Topic 3: Operators
  - Topic 4: Control Structures, Loop & Comprehension

Day 2

- Day 2 — Functions, Modules & Packages
  - Day 1 recap
  - Topic 5: Function
  - Lunch (30 min)
  - Topic 6: Modules & Packages
  - Course feedback, wrap-up and Q&A
  - Daily timing: 9:30am – 5:30pm · 30-min lunch

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WHAT WE'LL COVER

Course Outline — 6 Topics · 21 Hands-On Activities

1

Topic 1 · Get Started on Python

Overview · install · IDE · first script · comments

2

Topic 2 · Data Types

Number · String · List · Tuple · Dictionary · Set

3

Topic 3 · Operators

Arithmetic · compound · comparison · membership · logical

4

Topic 4 · Control Structures

Conditional · loops · multiple sequences · comprehension

5

Topic 5 · Function

Syntax · returns · arguments · lambda, map, filter

6

Topic 6 · Modules & Packages

Standard library · pip · pandas, matplotlib, yfinance

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01

TOPIC 01

Get Started on Python

Overview of Python · Install Python · Python IDE · First Script · Comments

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KEY CONCEPTS

Key Concepts — Get Started on Python

1

Powerful & multi-purpose

Python is used in data science, AI, finance, web and automation.

2

Simple, readable syntax

Easy to learn — the ideal first programming language.

3

Interpreted language

Code runs line by line — no compile step needed.

4

Object-oriented

Everything in Python is an object.

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OVERVIEW

What is Python?

1

Powerful & multi-purpose

Used in data science, AI, web, finance, engineering and automation.

2

Simple, easy-to-use syntax

Reads almost like English — perfect for beginners.

3

Interpreted language

Run your code immediately, line by line — no compiling.

4

Object-oriented

Everything in Python is an object.

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THE #1 LANGUAGE

Why Python?

1

Ranked #1

Top programming language for years running (IEEE Spectrum).

2

Data science & machine learning

The default language of analytics, ML and AI.

3

Finance & engineering

Widely used for financial modelling and engineering computation.

4

Huge ecosystem

Hundreds of thousands of free packages on PyPI.

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A SHORT TIMELINE

History of Python

1. 1989 — Guido van Rossum starts Python
2. 1991 — First public release
3. 2000 — Python 2.0 released
4. 2008 — Python 3.0 released
5. Today — Python 3.x, the industry standard

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SETUP

Download and Install Python

1. Go to www.python.org/downloads
2. Download the installer for Windows / macOS
3. Tick 'Add Python to PATH', then Install
4. Verify in a terminal: python --version

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WHERE YOU'LL CODE

Choose Your Python IDE

1

Google Colab

Cloud-based, zero install — convenient; we use it in class.

2

Jupyter Notebook / JupyterLab

Popular with data analysts.

3

Visual Studio Code

Popular with developers.

4

PyCharm

Full-featured professional Python IDE.

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CLOUD IDE

Run Python on Google Colab

1. Open colab.research.google.com
2. Sign in with a Google account
3. File ▸ New notebook
4. Type code in a cell
5. Press Shift+Enter to run

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HELLO WORLD

Your First Python Script

- All Python script files are saved with a .py extension.
- Run a script from the terminal / CMD prompt.
- In Colab or Jupyter, just press Shift+Enter.

python

```
print("Hello World!")

# save as test.py, then run:
#   python test.py
```

OUTPUT

```
Hello World!
```


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DOCUMENT YOUR CODE

Python Comments

- # starts a single-line comment.
- Triple quotes """ … """ wrap multi-line comments.
- Comments explain the why — Python ignores them.

python

```python
# this is a single-line comment
"""
This is a
multi-line comment
"""
print("Comments make code readable")
```

OUTPUT

```
Comments make code readable
```


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TOPIC 01 · HANDS-ON

Activity: Your First Python Script

ACTIVITY 1

Python coding is a lot of fun and easy to start. Set up your IDE (Google Colab, Visual Studio Code or another IDE), run your first Python script and add comments.

You'll build

A working Python environment and a script that prints output with comments.

Tools: Google Colab / VS Code, Python 3 **Step-by-step guide:** **Learner Guide**

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ACTIVITY 1 · VERIFY

Activity: Your First Python Script

✅ Check your result

The notebook / terminal prints your messages exactly once per print() call, and the commented lines produce no output.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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TOPIC RECAP

Recap — Get Started on Python

- Activity 1: Your First Python Script — A working Python environment and a script that prints output with comments.

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02

TOPIC 02

Data Types

Number · String · List · Tuple · Dictionary · Set

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KEY CONCEPTS

Key Concepts — Data Types

1

Six major data types

Number, String, List, Tuple, Dictionary and Set.

2

Mutable vs immutable

Lists change in place; tuples and strings never change.

3

Formatting output

format() and f-strings shape numbers and text for reports.

4

Build on the basics

Stacks, queues, trees and graphs are built from these six.

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THE BUILDING BLOCKS

Six Major Data Types

1

Number

int, float, complex — 7, 3.14, 2+3j

2

String

Text in quotes — 'Hello'

3

List

Ordered, mutable — [1, 2, 3]

4

Tuple

Ordered, immutable — (1, 2, 3)

5

Dictionary

key : value pairs — {'a': 1}

6

Set

Unordered, distinct — {1, 2, 3}

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NUMBER

Numbers & Type Inference

- Python supports integers, floating-point and complex numbers.
- Type inference: Python works out the type for you.
- Check any type with type().

python

```python
a = 7          # int
b = 3.14       # float
c = 2 + 3j     # complex
print(type(a), type(b), type(c))
```

OUTPUT

```
<class 'int'> <class 'float'> <class 'complex'>
```


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NUMBER

Multiple Assignment

- Assign values to multiple variables in one line.
- Both sides are evaluated first, then assigned together —
- this matters later for the Fibonacci lab!

python

```python
x, y = 1, 2
x, y = y, x + y   # simultaneous update
print(x, y)
```

OUTPUT

```
2 3
```


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PYTHON RULE

Indentation Matters

- Python forces proper indentation — it defines the code block.
- All lines in the same block must share the same indent.
- Inconsistent spacing raises an IndentationError.

python

```
if True:
    print("inside the block")
      print("oops")   # IndentationError!
```

OUTPUT

```
IndentationError: unexpected indent
```


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Formatting Numbers — format()

- {} is the placeholder for each value.
- {:10.2f} → width 10 characters, 2 decimal places.
- Great for lining up financial figures.

python

```
amt = 1234.5678
print("Amount: {:10.2f}".format(amt))
print("Rate:   {:.1%}".format(0.0525))
```

OUTPUT

```
Amount:    1234.57
Rate:   5.2%
```


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MODERN & POPULAR

Formatting with f-Strings

- Add an f before the quote: f"…{variable}…".
- Put format specs after a colon: {amt:.2f}.
- f-strings are the modern, preferred style.

python

```
amt = 1234.5678
print(f"Amount: {amt:.2f}")
name = "budget"
print(f"A {name} is a quantitative plan")
```

OUTPUT

```
Amount: 1234.57
A budget is a quantitative plan
```


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Activity: Format Numbers

ACTIVITY 2

Given a = 4.444, b = 5.555 and c = 6.6666, use the string-literal (f-string) method to format the output as: 5.55 + 6.67 + 4.4 = 17

You'll build

An f-string that rounds each number to the required decimal places.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** Learner Guide

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ACTIVITY 2 · VERIFY

Activity: Format Numbers

✅ Check your result

The output is exactly: 5.55 + 6.67 + 4.4 = 17

Follow the detailed step-by-step walkthrough in the Learner Guide.

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READABLE CODE

Multiline Statements

- Long lines can be split for readability.
- Use a backslash \ as the line-continuation character.
- Inside ( ), [ ] or { } you can break lines freely.

python

```
total = 1 + 2 + 3 + \
        4 + 5 + 6
print(total)
```

OUTPUT

```
21
```


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String

- Single or double quotes — but be consistent.
- + joins (concatenates) strings.
- * repeats a string.

python

```
a = 'Hello'
b = " World"
print(a + b)
print(a * 2)
```

OUTPUT

```
Hello World
HelloHello
```


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STRING TOOLBOX

String Methods

1. **c.upper()**
   ALL CAPITALS

2. **c.lower()**
   all lowercase

3. **c.capitalize()**
   First letter capital

4. **c.replace(a, b)**
   Swap a for b

5. **c.strip()**
   Trim whitespace

6. **c.split(sep)**
   Break into a list

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TOPIC 02 · HANDS-ON

Activity: Format String

ACTIVITY 3

Given bs = 'balance sheet' and income = 'income statement', print: "The finance statements typically require Balance sheet and INCOME STATEMENT." using the string-literal (f-string) method.

You'll build

An f-string combining capitalize() and upper() string methods.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 3 · VERIFY

Activity: Format String

✅ Check your result

The output is exactly: "The finance statements typically require Balance sheet and INCOME STATEMENT."

Follow the detailed step-by-step walkthrough in the Learner Guide.

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STRING

Strip · Split · Join

- strip() removes leading/trailing whitespace.
- split(sep) breaks a string into a list.
- sep.join(list) glues a list back into a string.

python

```
email = 'finance@company.com'
user, domain = email.split('@')
print(user, domain)
print('/'.join([user, domain]))
```

OUTPUT

```
finance company.com
finance/company.com
```


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TOPIC 02 · HANDS-ON

Activity: Split and Join Strings

ACTIVITY 4

Given email = 'finance@company.com', separate the user name and domain name, then join them back together with a '/'.

You'll build

Code that splits an email address and re-joins the parts with a different separator.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** Learner Guide

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ACTIVITY 4 · VERIFY

Activity: Split and Join Strings

✅ Check your result

The output shows 'finance', then 'company.com', then 'finance/company.com'.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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List & Slicing

- An ordered collection of mutable items of any type.
- Enclosed by [square brackets].
- Slice with [start:stop:step]; negatives count from the end.

python

```
a = [7, 8, 9, 10, 11]
print(a[0], a[-1])
print(a[0:3])
print(a[0:4:2])
```

OUTPUT

```
7 11
[7, 8, 9]
[7, 9]
```


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LIST TOOLBOX

List Methods

1. **append(x) / extend(y)**
   Add one item / add a whole list

2. **insert(i, x)**
   Insert x at position i

3. **remove(x) / pop(i)**
   Remove by value / by position

4. **index(x) / count(x)**
   Find position / count occurrences

5. **sort() / reverse()**
   Order the list / flip the order

6. **copy()**
   Duplicate the list safely

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TOPIC 02 · HANDS-ON

Activity: List Operations

ACTIVITY 5

Given years = ['2017','2018','2020','2021']: copy it to years2, remove '2021' from years2, insert '2019' at position 2, then reverse the order of years2.

You'll build

A list transformed step by step with the core list methods.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 5 · VERIFY

Activity: List Operations

✅ Check your result

years is unchanged ['2017','2018','2020','2021'] and years2 prints ['2020','2019','2018','2017'].

Follow the detailed step-by-step walkthrough in the Learner Guide.

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KNOW THE DIFFERENCE

Tuple vs List

Tuple (parentheses)

- Immutable — cannot be changed
  - t = (1, 2, 3)
  - Safer for fixed data
  - Slicing works the same way

List [brackets]

- Mutable — add, remove, sort
  - a = [1, 2, 3]
  - Rich methods: append, pop, sort…
  - Use when the data will change

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KEY : VALUE

Dictionary

- A collection of keys, each mapped to a value.
- Enclosed by { curly braces }, key separated by a colon.
- Methods: keys(), values(), pop(key), del, len().

python

```
px = {'DBS': 35.2, 'UOB': 29.8, 'OCBC': 13.1}
print(px['DBS'])
print(list(px.keys()))
px.pop('UOB')
print(len(px))
```

OUTPUT

```
35.2
['DBS', 'UOB', 'OCBC']
2
```


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DISTINCT ITEMS

Set & Set Operations

1

Set basics

Unordered collection of distinct objects in { braces }

2

x.union(y)

All unique items from both sets

3

x.intersection(y)

Items common to both sets

4

x.difference(y)

In x but not in y

5

y.issubset(x)

True if every item of y is in x

6

x.issuperset(y)

True if x contains all of y

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TOPIC 02 · HANDS-ON

Activity: Set — Count Unique Vocabulary

ACTIVITY 6

Given the sentence 'A budget is a quantitative plan for acquiring and using resources over a specified period.', find the number of unique vocabulary words.

You'll build

A set-based word counter that removes duplicate words.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 6 · VERIFY

Activity: Set — Count Unique Vocabulary

✅ Check your result

The duplicate 'a' appears only once in the set, and the unique word count prints 13.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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REMEMBER THE BRACKETS

Summary of Data Types

1

List — [Bracket]
Ordered and mutable.

2

Tuple — (Parentheses)
Ordered and immutable.

3

Dictionary — { key: value }
Keys mapped to values.

4

Set — { Braces }
Unordered, distinct items.

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TOPIC RECAP

Recap — Data Types

- Activity 2: Format Numbers — An f-string that rounds each number to the required decimal places.
- Activity 3: Format String — An f-string combining capitalize() and upper() string methods.
- Activity 4: Split and Join Strings — Code that splits an email address and re-joins the parts with a different separator.
- Activity 5: List Operations — A list transformed step by step with the core list methods.
- Activity 6: Set — Count Unique Vocabulary — A set-based word counter that removes duplicate words.

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03

TOPIC 03

Operators

Arithmetic · Compound · Comparison · Membership · Logical

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KEY CONCEPTS

Key Concepts — Operators

1

Arithmetic & compound

+ - * / ** % // and their compound forms (+=, -=, …).

2

Comparison

< <= > >= == != compare values and return True/False.

3

Membership

in and not in test whether an item is inside a sequence.

4

Logical

and, or, not combine conditions into business rules.

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DO THE MATH

Arithmetic & Compound Operators

Arithmetic

- `+` addition
- `-` subtraction
- `*` multiplication
- `/` division
- `**` exponent
- `%` modulus · `//` floor division

Compound (x = x + 1 → x += 1)

- `+=` add and assign
- `-=` subtract and assign
- `*=` multiply and assign
- `/=` divide and assign
- `**=` exponent and assign
- `%=` · `//=` modulus / floor

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TRUE OR FALSE

Comparison & Membership Operators

Comparison

- < strictly less than
- <= less than or equal
- > strictly greater than
- >= greater than or equal
- == equal
- != not equal

Membership

- in — item is inside a sequence
- not in — item is absent
  - 'DBS' in ['DBS', 'UOB'] → True
  - 5 not in [1, 2, 3] → True

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COMBINE CONDITIONS

Logical (Boolean) Operators

- and — True only when both sides are True.
- or — True when at least one side is True.
- not — flips True to False and back.

python

```
a = 1
b = 2
print(a == 1 and b > 1)
print(a > 1 or b > 1)
print(not (b > 3))
```

OUTPUT

```
True
True
True
```


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Activity: Operators

ACTIVITY 7

Given a = 1 and b = 2, predict and then verify: what is (a == 1) and not (b > 3)? Then explore the arithmetic and membership operators.

You'll build

Verified predictions of Boolean expressions and arithmetic results.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 7 · VERIFY

Activity: Operators

✅ Check your result

The Boolean expression prints True; 2**10 → 1024, 7%3 → 1, 7//3 → 2, x → 12, and both membership checks print True.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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TOPIC RECAP

Recap — Operators

- Activity 7: Operators — Verified predictions of Boolean expressions and arithmetic results.

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04

TOPIC 04

Control Structures, Loop and Comprehension

Conditional · While & For Loops · Multiple Sequences · Comprehension

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KEY CONCEPTS

Key Concepts — Control Structures, Loop and Comprehension

1

Conditional

if / elif / else route the program down different paths.

2

Loops

while repeats until a condition fails; for iterates a sequence.

3

Multiple sequences

enumerate() and zip() loop several lists together.

4

Comprehension

One-line list / set / dict builders: [transform iteration filter].

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CONDITIONAL

If – Elif – Else

- if tests a condition; elif tests the next; else catches the rest.
- The condition line ends with a colon :
- The block under each branch is indented.

python

```
stock_price = 299
if stock_price > 300:
    print("Sell")
elif stock_price == 300:
    print("Hold")
else:
    print("Buy")
```

OUTPUT

```
Buy
```


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ONE-LINE IF

Ternary Operator

- A compact if/else on a single line.
- value_if_true if condition else value_if_false

python

```
order = 200
discount = 25 if order > 100 else 0
print(discount)
```

OUTPUT

```
25
```


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TOPIC 04 · HANDS-ON

Activity: Conditional — Income Classifier

ACTIVITY 8

Prompt the user for their income and classify it: more than 8000 → 'High income'; 4000 to 8000 → 'Medium high income'; otherwise → 'Average income'.

You'll build

An interactive income classifier using input() and if/elif/else.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 8 · VERIFY

Activity: Conditional — Income Classifier

✅ Check your result

9000 → High income, 5000 → Medium high income, 3000 → Average income; typing 'abc' re-prompts instead of crashing.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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HANDLE BAD INPUT

Robust Input — try / except

- int(input()) crashes if the user types text like 'abc'.
- Wrap risky code in try: … except ValueError: …
- Combine with a while True loop to re-prompt until valid.

python

```python
while True:
    try:
        age = int(input('What is your age? '))
        break
    except ValueError:
        print('Invalid input. Enter a number.')
print('Age:', age)
```

OUTPUT

```
What is your age? abc
Invalid input. Enter a number.
What is your age? 42
Age: 42
```


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LOOP UNTIL

While Loop

- A while loop = a condition + a block of code.
- The block repeats as long as the condition stays True.
- Update the loop variable inside the block — or loop forever!

python

```
n = 1
while n <= 5:
    print(n, end=' ')
    n += 1
```

OUTPUT

```
1 2 3 4 5
```


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MATH IN NATURE, ART & FINANCE

Fibonacci & the Golden Ratio

1

The sequence

1, 1, 2, 3, 5, 8, 13, 21, 34 … each number is the sum of the previous two.

2

In nature

Sunflower spirals, shells and petals follow Fibonacci counts.

3

In art & architecture

The golden ratio (1.618…) shapes classic compositions.

4

In trading

Fibonacci retracement levels are used in technical analysis of security prices.

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TOPIC 04 · HANDS-ON

Activity: Fibonacci Sequence & Golden Ratio (While Loop)

ACTIVITY 9

A Fibonacci number is the sum of the previous two: 1, 1, 2, 3, 5, 8, 13, 21, 34 … Use a while loop to compute the sequence up to 144 and compute the golden ratio.

You'll build

A while loop producing the Fibonacci sequence and the golden ratio ≈ 1.618.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** Learner Guide

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ACTIVITY 9 · VERIFY

Activity: Fibonacci Sequence & Golden Ratio (While Loop)

✅ Check your result

The sequence prints 1 1 2 3 5 8 13 21 34 55 89 144 and the golden ratio is approximately 1.618.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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LOOP OVER A SEQUENCE

For Loop & range()

- for executes a block once per item in a sequence.
- range(start, stop, step) generates a number sequence.
- Question: what is list(range(2, 12, 2))?

python

```
for i in range(1, 6):
    print(i, end=' ')

print()
print(list(range(2, 12, 2)))
```

OUTPUT

```
1 2 3 4 5
[2, 4, 6, 8, 10]
```


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TOPIC 04 · HANDS-ON

Activity: For Loop — Square Numbers

ACTIVITY 10

Generate a Python list of 10 square numbers [1, 4, 9, 16, 25, 36, 49, 64, 81, 100] using a for loop and range().

You'll build

A list of the first 10 square numbers built inside a for loop.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 10 · VERIFY

Activity: For Loop — Square Numbers

✅ Check your result

The list prints [1, 4, 9, 16, 25, 36, 49, 64, 81, 100] and range(2,12,2) gives [2, 4, 6, 8, 10].

Follow the detailed step-by-step walkthrough in the Learner Guide.

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INDEX + VALUE TOGETHER

Enumerate

- enumerate() yields (index, value) pairs as you loop.
- Use the index to reach into a second list.

python

```
year = ['2017', '2018', '2019']
revenue = [150000, 200000, 500000]
for i, y in enumerate(year):
    print(f"{y}'s revenue is ${revenue[i]}")
```

OUTPUT

```
2017's revenue is $150000
2018's revenue is $200000
2019's revenue is $500000
```


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Zip

- zip() pairs up items from several sequences.
- Cleaner than indexing when lists move together.

python

```
year = ['2017', '2018', '2019']
revenue = [150000, 200000, 500000]
for y, r in zip(year, revenue):
    print(f"{y}'s revenue is ${r}")
```

OUTPUT

```
2017's revenue is $150000
2018's revenue is $200000
2019's revenue is $500000
```


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TOPIC 04 · HANDS-ON

Activity: Loop Multiple Sequences with zip()

ACTIVITY 11

Given year, revenue and expense lists, fill in the blank so the loop prints each year's revenue and expense together: data = _____; for y, r, e in data: …

You'll build

A zip() of three lists looped in parallel to print a mini financial report.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 11 · VERIFY

Activity: Loop Multiple Sequences with zip()

✅ Check your result

Three lines print, one per year, each showing the matching revenue and expense (e.g. 2017's revenue is $150000 and expense is $90000).

Follow the detailed step-by-step walkthrough in the Learner Guide.

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CONTROL THE LOOP

Break vs Continue

break — exit the loop

- Stops the whole loop at once

– for i in range(10):

– if i == 3: break

– → prints 0 1 2

continue — skip this round

- Jumps to the next iteration

– for i in range(5):

– if i == 3: continue

– → prints 0 1 2 4

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Comprehension — List, Set, Dict

- Python offers 3 comprehensions: list, set and dict.
- Pattern: [transform iteration filter]
- Same idea with { } for sets and {key: value} for dicts.

python

```
squares = [x**2 for x in range(1, 6)]
evens   = {x for x in range(10) if x % 2 == 0}
sqmap   = {x: x**2 for x in range(1, 4)}
print(squares)
print(evens)
print(sqmap)
```

OUTPUT

```
[1, 4, 9, 16, 25]
{0, 2, 4, 6, 8}
{1: 1, 2: 4, 3: 9}
```


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Activity: List Comprehension

ACTIVITY 12

Create a list comprehension of 10 square numbers that are not divisible by 3 and 5: [1, 4, 16, 49, 64, 121, 169, 196, 256, 289]

You'll build

A one-line comprehension with a transform, an iteration and a filter.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 12 · VERIFY

Activity: List Comprehension

✅ Check your result

The list prints [1, 4, 16, 49, 64, 121, 169, 196, 256, 289] with length 10.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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TOPIC RECAP

Recap — Control Structures, Loop and Comprehension

- Activity 8: Conditional — Income Classifier — An interactive income classifier using input() and if/elif/else.
- Activity 9: Fibonacci Sequence & Golden Ratio (While Loop) — A while loop producing the Fibonacci sequence and the golden ratio ≈ 1.618.
- Activity 10: For Loop — Square Numbers — A list of the first 10 square numbers built inside a for loop.
- Activity 11: Loop Multiple Sequences with zip() — A zip() of three lists looped in parallel to print a mini financial report.
- Activity 12: List Comprehension — A one-line comprehension with a transform, an iteration and a filter.

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DAY 2

Functions, Modules & Packages

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WARM-UP

Recap on Day 1 Topics

1

Recap quiz

Open the Recap Quiz from the LMS and try the questions.

2

Topic 1–2

Python setup, first script, six data types.

3

Topic 3–4

Operators, if/elif/else, loops and comprehension.

4

Review

We review the answers together before starting Day 2.

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05

TOPIC 05

Function

Function Syntax · Return Values · Default & Variable Arguments · Lambda, Map, Filter

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KEY CONCEPTS

Key Concepts — Function

1

Reusable blocks

def packages logic once; call it everywhere.

2

Return values

Functions can return one value — or several at once.

3

Flexible arguments

Default, named and variable (*args) arguments.

4

Lambda, map, filter

Small anonymous functions that transform and filter data.

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Function Syntax

- Functions are reusable blocks of code that perform specific tasks.
- def name(parameters): then an indented body.
- return sends the result back to the caller.

python

```
def add(x, y):
    z = x + y
    return z

print(add(3, 5))
```

OUTPUT

```
8
```


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Activity: Function — Future Value Calculator

ACTIVITY 13

The FV calculation lets investors predict the profit generated by an investment. Create a function to compute the future value FV = PV × (1 + r)^n for a present value PV, annual interest rate r and n years.

You'll build

A reusable fv() function returning the future value of an investment.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** Learner Guide

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ACTIVITY 13 · VERIFY

Activity: Function — Future Value Calculator

✅ Check your result

fv(10000, 0.05, 10) prints approximately 16288.95 — formatted as $16,288.95.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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MORE THAN ONE ANSWER

Return Multiple Values

- Separate the return values with commas.
- Unpack them into multiple variables on the left.

python

```
def f(x):
    return x, 2 * x

a, b = f(5)
print(a, b)
```

OUTPUT

```
5 10
```


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Activity: Multiple Return Values — Discount & Tax

ACTIVITY 14

Prompt the user for the order amount, then create a function that returns BOTH the discount amount and the tax for a grocery purchase: discount is 25% if order > 200 else 0; disc_amt = discount*order/100; tax = 7% of the discounted amount.

You'll build

A function returning (discount amount, tax) unpacked into two variables.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 14 · VERIFY

Activity: Multiple Return Values — Discount & Tax

✅ Check your result

Order 300 → Discount $75.00 and Tax $15.75; order 100 → Discount $0.00 and Tax $7.00.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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FLEXIBLE CALLS

Default & Named Arguments

- Default arguments supply a value when the caller omits one.
- Named arguments can be passed in any order.
- Question: def f(a=3, b=5): return a*b — what is f(a=4)?

python

```python
def f(a=3, b=5):
    return a * b

print(f())         # both defaults
print(f(b=8))      # named
print(f(b=6, a=3))
print(f(a=4))
```

OUTPUT

```
15
24
18
20
```


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ANY NUMBER OF INPUTS

Variable Arguments — *args

- *args collects any number of positional arguments into a tuple.
- Loop over args inside the function.

python

```python
def total(*args):
    s = 0
    for x in args:
        s += x
    return s

print(total(1, 2, 3))
print(total(5, 4, 7, 9))
```

OUTPUT

```
6
25
```


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Activity: Variable Arguments — Minimum Finder

ACTIVITY 15

Given a variable number of elements, create a function to determine the minimum value. E.g. for 5, 4, 7 the function returns 4. Do it with a loop — no min()!

You'll build

A my_min(*args) function that scans any number of values for the smallest.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** Learner Guide

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ACTIVITY 15 · VERIFY

Activity: Variable Arguments — Minimum Finder

✅ Check your result

my_min(5, 4, 7) prints 4 and my_min(9, 2, 8, 1, 6) prints 1 — matching min().

Follow the detailed step-by-step walkthrough in the Learner Guide.

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SMALL & ANONYMOUS

Lambda Function

- lambda creates a small function without a name.
- Temporary — used right where it is created.
- Often passed into other functions like map() and filter().

python

```
f = lambda x, y: 10 * x + y
print(f(4, 2))

sq = lambda x: x ** 2
print(sq(6))
```

OUTPUT

```
42
36
```


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TOPIC 05 · HANDS-ON

Activity: Lambda Function

ACTIVITY 16

Given f = lambda x, y: 10*x + y and def g(f, x, y=0): return f(x, y) — what is g(f, 4)? Predict first, then verify.

You'll build

A verified trace of a lambda passed into another function with a default argument.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 16 · VERIFY

Activity: Lambda Function

✅ Check your result

g(f, 4) prints 40 and cube(4) prints 64.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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TRANSFORM EVERY ITEM

Map

- map(function, sequence) applies the function to every element.
- Works with lambda and with several sequences at once.
- Wrap in list() to see the results.

python

```
x = [1, 2, 3, 4]
print(list(map(lambda v: v ** 2, x)))

y = [10, 20, 30, 40]
print(list(map(lambda a, b: a + b, x, y)))
```

OUTPUT

```
[1, 4, 9, 16]
[11, 22, 33, 44]
```


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Activity: Map

ACTIVITY 17

Use map() to compute the answers for f = lambda x, y: 10*x + y with x = [0, 1, 2, 3] and y = [2, 4, 6, 8].

You'll build

A map() over two lists producing [2, 14, 26, 38].

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 17 · VERIFY

Activity: Map

✅ Check your result

The result prints [2, 14, 26, 38].

Follow the detailed step-by-step walkthrough in the Learner Guide.

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KEEP WHAT PASSES THE TEST

Filter

- filter(function, list) keeps items where the function is True.
- An elegant way to select the elements you need.

python

```
x = [0, 1, -4, 2, -6, 3]
pos = list(filter(lambda v: v >= 0, x))
print(pos)
```

OUTPUT

```
[0, 1, 2, 3]
```


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TOPIC 05 · HANDS-ON

Activity: Filter

ACTIVITY 18

Use filter() to remove all the negative values in x = [0, 1, -4, 2, -6, 3].

You'll build

A filter() with a lambda that keeps only non-negative numbers.

Tools: Python 3 (Colab / IDE) **Step-by-step guide:** Learner Guide

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ACTIVITY 18 · VERIFY

Activity: Filter

✅ Check your result

Both approaches print [0, 1, 2, 3] — the negatives are gone.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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TOPIC RECAP

Recap — Function

- Activity 13: Function — Future Value Calculator — A reusable fv() function returning the future value of an investment.
- Activity 14: Multiple Return Values — Discount & Tax — A function returning (discount amount, tax) unpacked into two variables.
- Activity 15: Variable Arguments — Minimum Finder — A my_min(*args) function that scans any number of values for the smallest.
- Activity 16: Lambda Function — A verified trace of a lambda passed into another function with a default argument.
- Activity 17: Map — A map() over two lists producing [2, 14, 26, 38].
- Activity 18: Filter — A filter() with a lambda that keeps only non-negative numbers.

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06

TOPIC 06

Modules & Packages

Import Modules · Standard Library · Third-Party Packages · pip · Data & Charts

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KEY CONCEPTS

Key Concepts — Modules & Packages

1

Standard library

math, time, random, datetime — batteries included.

2

import

import module, import module as alias, from module import name.

3

Third-party packages

pip installs from PyPI: numpy, pandas, matplotlib, yfinance.

4

Real finance data

Load CSV / Yahoo Finance data and visualize stock prices.

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BATTERIES INCLUDED

Python Standard Library

1

math

pi, e, sin, cos, log, sqrt, ceil, floor …

2

time

time.sleep(secs) and time functions.

3

random

random.choice([1,2,3,4,5,6]) — roll a die.

4

Explore more

Full index: docs.python.org/3/library

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THREE WAYS TO IMPORT

Import Modules

- import module — call with the module prefix.
- import module as alias — shorter prefix.
- from module import * — no prefix (use sparingly).

python

```
import math
print(math.sin(0.5))

import math as m
print(m.sqrt(16))

from math import pi
print(pi)
```

OUTPUT

```
0.479425538604203
4.0
3.141592653589793
```


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STANDARD MODULES IN ACTION

math · time · random

- math for maths functions, constants and rounding.
- time.sleep() pauses your script.
- random picks random values — simulations and games.

python

```
import math, random, time
print(math.pi, math.ceil(10.3))
die = random.choice([1, 2, 3, 4, 5, 6])
print('You rolled', die)
time.sleep(1)   # pause 1 second
```

OUTPUT

```
3.141592653589793 11
You rolled 4
```


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THE PYTHON SUPERPOWER

Third-Party Packages

1

PyPI

Find every third-party package at pypi.org.

2

pip install

pip install numpy pandas matplotlib yfinance

3

Anaconda

150+ pre-built scientific packages in one install — anaconda.com.

4

In Colab

Most data packages are already installed for you.

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TOPIC 06 · HANDS-ON

Activity: Install Third-Party Packages

ACTIVITY 19

Install the course packages on your terminal / CMD prompt (or verify them in Colab): numpy, matplotlib, pandas and yfinance — then import them to confirm.

You'll build

A working data-analysis toolkit: numpy, pandas, matplotlib and yfinance.

Tools: pip, PyPI, Google Colab / terminal **Step-by-step guide:** **Learner Guide**

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ACTIVITY 19 · VERIFY

Activity: Install Third-Party Packages

✅ Check your result

All four packages import without ModuleNotFoundError and print a version number.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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Import Stock Data from a CSV File

- pandas reads tabular data with one line.
- head() previews the first rows.
- Select a column with df[['Close']].

python

```
import pandas as pd
stock = pd.read_csv('singtel.csv')
print(stock.head())
stock_close = stock[['Close']]
```

OUTPUT

```
        Date   Open   High    Low  Close
0 2024-01-02  2.46   2.48   2.44   2.47
...
```


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Visualize Stock Price History

- matplotlib.pyplot draws the chart.
- plot() the closing price, add title and axis labels.
- show() displays the figure.

python

```
import matplotlib.pyplot as plt
plt.figure(figsize=(14, 5))
plt.plot(stock_close, 'b')
plt.plot(stock_close, 'ro')
plt.title('Singtel Close Price')
plt.xlabel('Trading Days')
plt.ylabel('Close Price')
plt.show()
```


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TOPIC 06 · HANDS-ON

Activity: Import & Visualize Stock Data from CSV

ACTIVITY 20

Import historical stock data from a CSV file with pandas, select the closing price and visualize the price history with matplotlib.

You'll build

A line chart of the Singtel closing-price history from a CSV file.

Tools: pandas, matplotlib, CSV (labs/data/singtel.csv) **Step-by-step guide:** **Learner Guide**

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ACTIVITY 20 · VERIFY

Activity: Import & Visualize Stock Data from CSV

✅ Check your result

head() shows the CSV columns (Date, Open, High, Low, Close) and a blue line chart with red markers appears, titled 'Singtel Close Price Representation'.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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YFINANCE

Import Data from Yahoo Finance

- yfinance downloads live market data.
- Pick a ticker and a date range.
- Plot the Close column just like the CSV data.

python

```
import yfinance as yf
stock = yf.download('Z74.SI',
        start='2024-01-01', end='2024-12-31')
stock_close = stock[['Close']]
print(stock_close.tail())
```

OUTPUT

```
              Close
Date
2024-12-27    3.10
...
```


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TOPIC 06 · HANDS-ON

Activity: Import Data from Yahoo Finance

ACTIVITY 21

Use the yfinance package to download Singtel (Z74.SI) stock data from Yahoo Finance for a date range, then plot the closing price.

You'll build

A live Yahoo Finance download and a closing-price chart.

Tools: yfinance, pandas, matplotlib **Step-by-step guide:** Learner Guide

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ACTIVITY 21 · VERIFY

Activity: Import Data from Yahoo Finance

✅ Check your result

tail() shows recent trading rows and the chart plots a year of closing prices; swapping the ticker redraws the chart for the new stock.

Follow the detailed step-by-step walkthrough in the Learner Guide.

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PUT IT ALL TOGETHER

Practice Session

1

Scenario-based challenge

Open the Practice Section on the LMS — a real business scenario.

2

Code it yourself

Apply data types, loops, functions and packages together.

3

Discuss

Compare approaches with the class.

4

Ask

The trainer is here to help — no question is too small.

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TOPIC RECAP

Recap — Modules & Packages

- Activity 19: Install Third-Party Packages — A working data-analysis toolkit: numpy, pandas, matplotlib and yfinance.
- Activity 20: Import & Visualize Stock Data from CSV — A line chart of the Singtel closing-price history from a CSV file.
- Activity 21: Import Data from Yahoo Finance — A live Yahoo Finance download and a closing-price chart.

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WRAP-UP

Course Summary & Next Steps

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DISCUSSION

How would you use Python in your workplace?

Share one task you will automate or analyse with Python — reports, data cleaning, dashboards, market data …

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COURSE SUMMARY

What You Achieved

1

Python foundations

Set up Python, ran scripts, mastered the six data types.

2

Operators & control flow

Built business rules with operators, conditionals, loops and comprehensions.

3

Functions

Packaged logic into reusable functions — including lambda, map and filter.

4

Modules & packages

Imported standard modules and installed pandas, matplotlib and yfinance.

5

Real finance data

Loaded CSV and Yahoo Finance data and visualized stock prices.

6

21 hands-on activities

Every topic practised with step-by-step labs in the Learner Guide.

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WRAP-UP

Summary & Q&A

- Six topics: getting started, data types, operators, control structures, functions, modules & packages.
- The Learner Guide carries the detailed step-by-step of all 21 activities — keep practising with it.
- Any remaining questions before we wrap up?

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CONTINUE YOUR LEARNING JOURNEY

Recommended Next Courses

1. **Python Data Analysis Masterclass**
   Hands-on course · Tertiary Courses

2. **AI Vibe Coding with Python**
   Hands-on course · Tertiary Courses

3. **SQL Masterclass**
   Hands-on course · Tertiary Courses

4. **AI Vibe Coding for Machine Learning**
   Hands-on course · Tertiary Courses

5. **AI Vibe Coding for Deep Learning**
   Hands-on course · Tertiary Courses

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FEEDBACK & CERTIFICATE

Course Feedback & Certificate

- Share your feedback in the course survey on the LMS — it helps us improve the course.
- Your certificate of completion is issued via the LMS after the course.
- Portal: https://lms-tms.tertiaryinfotech.com/

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WE'RE HERE TO HELP

Support

- If you have any enquiries during and after the class, contact us:
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- Website: www.tertiarycourses.com.sg

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HAPPY CODING

Thank You!

You are now ready to code, test and improve Python programs for real business data.

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